concentrations on either side of the membrane also remain steady. Why? The resting potential arises from the net movement of far fewer ions than would be required to alter the concentration gradients.

If Na^+ is allowed to cross the membrane more readily, the membrane potential will move toward E_{Na} and away from E_K. As you'll see, this happens in generation of a nerve impulse.

Mastering Biology **Animation: Membrane Potentials**

CONCEPT CHECK 48.2

1. Under what circumstances could ions flow through an ion channel from a region of lower ion concentration to a region of higher ion concentration?
2. **WHAT IF?** Suppose a cell's membrane potential shifts from −70 mV to −50 mV. What changes in the cell's permeability to K^+ or Na^+ could cause such a shift?
3. **MAKE CONNECTIONS** Review Figure 7.11, which illustrates the diffusion of dye molecules across a membrane. Could diffusion eliminate the concentration gradient of a dye that has a net charge? Explain.

For suggested answers, see Appendix A.

CONCEPT 48.3

Action potentials are the signals conducted by axons

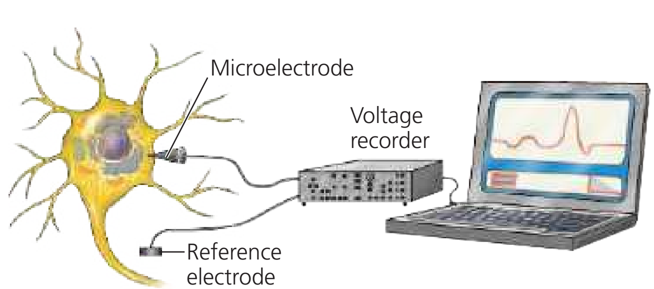
When a neuron responds to a stimulus, the membrane potential changes. Using intracellular recording, researchers can monitor these changes as a function of time **(Figure 48.9)**. As you will see, such recordings have been central to the study of information transfer by neurons.

▼ Figure 48.9 Research Method

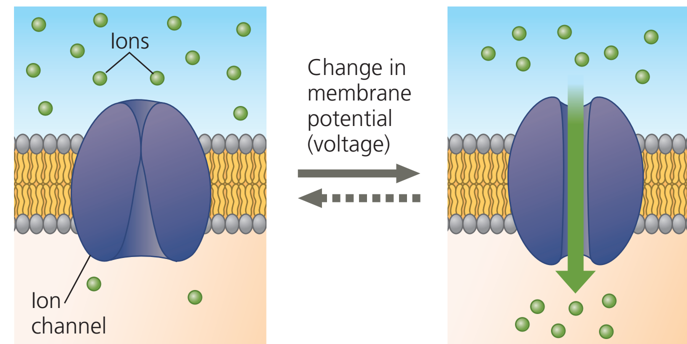
Intracellular Recording

Application Electrophysiologists use intracellular recording to measure the membrane potential of neurons and other cells.

Technique A microelectrode is made from a glass capillary tube filled with an electrically conductive salt solution. One end of the tube tapers to an extremely fine tip (diameter < 1 µm). While looking through a microscope, the experimenter uses a micropositioner to insert the tip of the microelectrode into a cell. A voltage recorder (usually an oscilloscope or a computer-based system) measures the voltage between the microelectrode tip inside the cell and a reference electrode placed in the solution outside the cell.



▼ Figure 48.10 Voltage-gated ion channel. A change in the membrane potential in one direction (solid arrow) opens the voltage-gated channel. The opposite change (dotted arrow) closes the channel.



Gate closed: No ions flow across membrane.

Gate open: Ions flow through channel.

VISUAL SKILLS *Gated ion channels allow ion flow in either direction. Using visual information in this figure, explain why there is net ion movement when the channel opens.*

How does a stimulus alter the membrane potential? Certain ion channels in a neuron, called **gated ion channels**, open or close in response to stimuli. When a gated ion channel opens or closes, it alters the membrane's permeability to particular ions **(Figure 48.10)**. The result is a rapid flow of ions across the membrane, altering the membrane potential.

Particular types of gated channels respond to different stimuli. For example, Figure 48.10 illustrates a **voltage-gated ion channel**, a channel that opens or closes in response to a shift in the voltage across the plasma membrane of the neuron. Later in this chapter, we will discuss gated channels that are located in neurons and are regulated by chemical signals.

Hyperpolarization and Depolarization

Let's consider now what happens in a neuron when a stimulus causes closed voltage-gated ion channels to open. If gated potassium channels in a resting neuron open, the membrane's permeability to K^+ increases. As a result, net diffusion of K^+ out of the neuron increases, shifting the membrane potential toward E_K (−90 mV at 37°C). This increase in the magnitude of the membrane potential, called a **hyperpolarization**, makes the inside of the membrane more negative **(Figure 48.11a)**. In a resting neuron, hyperpolarization results from any stimulus that increases the outflow of positive ions or the inflow of negative ions.

Although opening potassium channels in a resting neuron causes hyperpolarization, opening some other types of ion channels has an opposite effect, making the inside of the membrane less negative **(Figure 48.11b)**. A reduction in the